Task 1 (CRAAP Source Audit)

- Goal: Filter through Perplexity articles to verify that background information is reliable and provides solid evidence for Criterion A.1.
- Format: Individual or paired research activity (10–15 mins).

Student Activity Worksheet

Task 2 (PEEL Sentence Drafting)

Task 2: PEEL Guided Drafting (Sentence Builders)

PEEL Practice sheet

Formative task

- **Goal:** Build the structure of the write-up sentence-by-sentence using prompt questions focused on context, evidence, cause & effect, and justification.
- **Format:** Scaffolded writing organizer (15–20 mins).

Prompt Questions & Sentence Starters:

1. Point (Context, Target Audience & Problem)

- **Question:** Who is affected by this issue, where are they located, and what is the core problem?
- **Sentence Starter:** *"In [location/setting], [target audience] face a major issue where [describe the design problem]..."*

2. Evidence (CRAAP Source Fact)

- **Question:** What fact or statistic from your Task 1 CRAAP source proves this problem is real?
- **Sentence Starter:** *"According to [Publisher/Author], [insert stat/fact from Task 1]..."*

3. Explanation (Cause & Effect / Need for Automated Sensing)

- **Question:** What condition causes this problem to occur, and why is an automated, real-time alert or detection system necessary to prevent negative impacts?

- **Sentence Starter:** *"This problem occurs because [cause/condition occurs], which leads to [negative effect/consequence]. Without an automated detection system to monitor conditions in real time, [target audience/staff] cannot..."*

4. Link (Justification & Community Impact)

- **Question:** Why is a localized sensing system necessary, and how does addressing this issue promote fairness or community wellbeing?
- **Sentence Starter:** *"A dedicated monitoring system is therefore essential because existing methods [limitation of general apps/forecasts], and this localized solution ensures fair access to [community benefit]..."*

Student Activity: Formulating Their Own Questions

Have students select **one question from each row** (Factual, Conceptual, Debatable) that specifically applies to their chosen community issue.

Example

Inquiry Type	PEOPLE (Community & Wellbeing)	PLANET (Environment & Resources)	SYSTEMS / PROGRESS (Tech & Innovation)
Factual (Data & Facts)	*What health/safety risks do [target audience] face due to [issue]?*	*What are the safe environmental thresholds/limits for [toxin/level]?*	*How does a [sensor type] convert physical inputs into electrical signals?*
Conceptual (Relationships & Impact)	*How does lack of real-time warning impact the daily lives of vulnerable residents?*	*Why do localized environmental conditions differ from city-wide forecasts?*	*How do automated alert systems change human response times during emergencies?*
Debatable (Ethics & Fairness)	*To what extent is access to real-time safety technology a fundamental human right?*	*Should communities prioritize low-cost tech solutions over long-term environmental cleanup?*	*Is relying on automated sensor warnings better than depending on human monitoring?*

Manual on robotics

Resource for teaching

Task Goal

> **Goal:** Analyze an existing real-world sensor product by examining its **Part(s)**, **Purpose**, and **Complexity**, and identify a key limitation that your low-cost Wokwi sensor design will aim to solve for your target community.

Featured Product: Commercial Ultrasonic Flood Warning Unit

(A commercial IoT sensor system installed along riverbanks or drainage canals to warn communities of rising water levels.)

Student Instructions:

Examine the commercial sensor unit below and complete each row to break down how it functions, who it serves, and its limitations.

Task sheet

Helpsheet before Summative

Task Goal

> **Goal:** Synthesize your research findings to clearly outline your **Target Audience**, key **Research Insights**, and a **Basic Product Concept** before moving on to detailed designs in Criterion B.

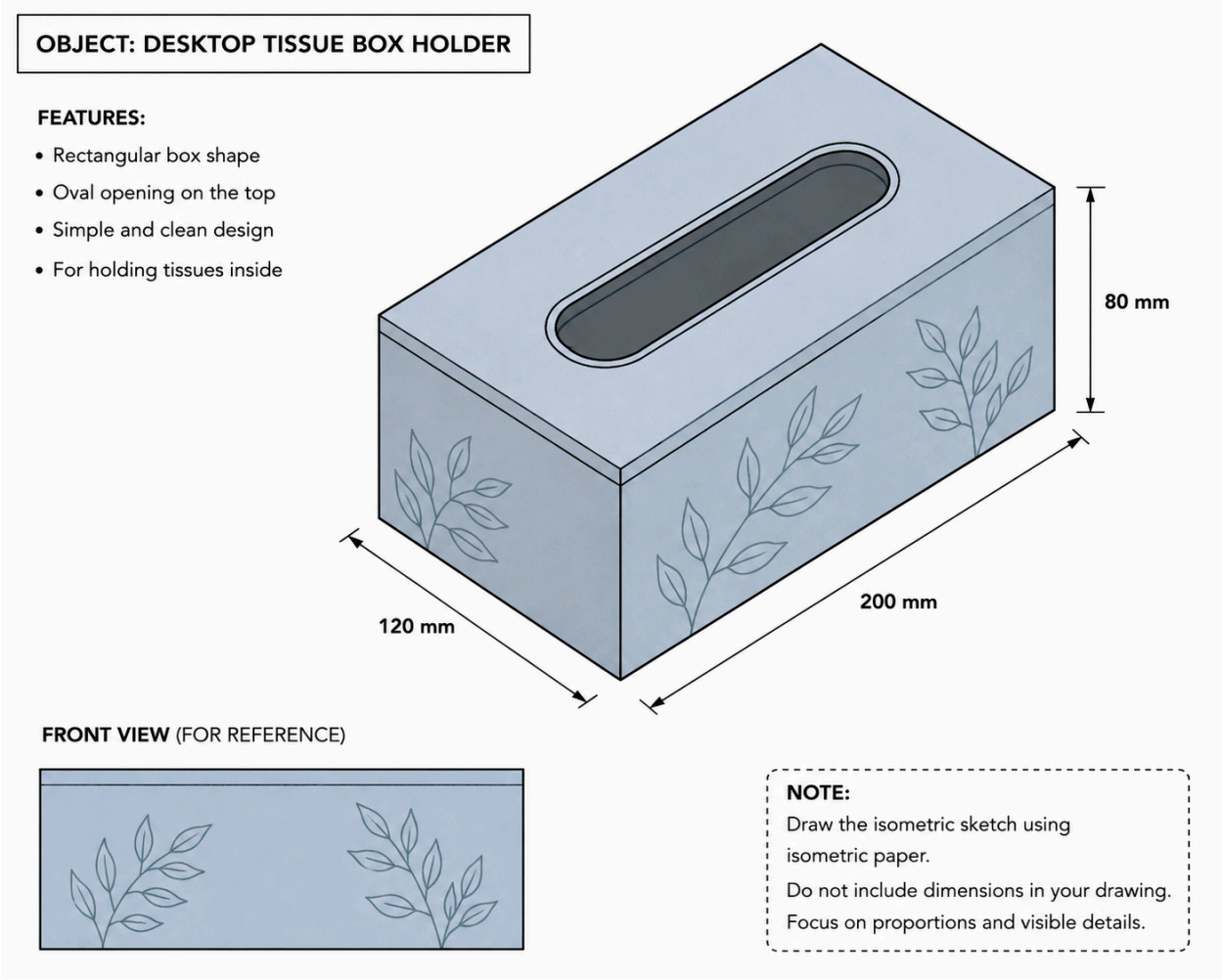
Complete the table below by summarizing your findings from your research, cause-and-effect analysis, and product analysis.

Tasksheet

Teaching Resource
Tasksheet

To be assessed as FA

In this classwork activity, you will learn the fundamentals of **isometric sketching** by watching the tutorial video. Apply the techniques demonstrated in the video to create a **freehand isometric sketch of a desktop tissue box holder** (attached here) on the



isometric paper. Focus on drawing accurate proportions, straight lines, and clearly representing the visible features of the product. This activity will help you develop the sketching skills needed to communicate your design ideas effectively in **Criterion B ii**.

Resources

- Watch the tutorial video shared by your teacher.
- Use isometric paper and a pencil.

Submission

- Complete one neat **freehand isometric sketch** of the desktop tissue box holder.
- Add your **name and class** to the sheet.
- Submit the completed sketch as an image

⭐ **Design Evaluation Matrix**

Criteria B – Strand iii

Instructions

Evaluate each design idea against your **design specifications**.

Use the following scoring scale:

- **0** = Does not meet the requirement
- **1** = Partly meets the requirement
- **2** = Mostly meets the requirement
- **3** = Fully meets the requirement

Design Specification	Idea 1	Idea 2	Idea 3
Functional Requirement	0 1 2 3	0 1 2 3	0 1 2 3
User & Safety	0 1 2 3	0 1 2 3	0 1 2 3
Sensor & Input	0 1 2 3	0 1 2 3	0 1 2 3
Output & Feedback	0 1 2 3	0 1 2 3	0 1 2 3
Sustainability & Community Impact	0 1 2 3	0 1 2 3	0 1 2 3
Practical to Build in Wokwi	0 1 2 3	0 1 2 3	0 1 2 3
Total Score (out of 18)	_____	_____	_____

My Final Choice

I have selected: ☐ Idea 1 ☐ Idea 2 ☐ Idea 3

Justification (4–5 sentences)

Explain why you selected this idea. Consider:

- Which idea best meets your design specifications?
- Which idea is most practical to build in Wokwi?
- Which idea best addresses the community problem?
- Which features from your other ideas could be improved or combined?
-

Answer:

Tasksheet

Students will be using this template.

IB MYP Design Student Workbook

Tasksheet

Canva Template

Strand 1 :

"Every evaluation question should help me determine whether one design specification has been achieved."
Scaffolding sheet - Designing Testing methods.

Strand 1 and 2 :

Tasksheet

Strand 3 and 4

Tasksheet

Students will use this template

MYP Design Story book_C and D

	Product Name	Product Image	Product Use
Product 1			
Product 2			
Product 3			

Scenario

Your design team has been asked to investigate why **single-use plastic water bottles continue to be widely used in schools despite reusable bottles being available.**

The school would like to design a product, campaign, or system that encourages students to make more responsible choices. Before developing a solution, you must first understand and analyse the problem using research.

Primary Research

A survey was conducted among **120 students**.

Survey Results

- **35%** bring a reusable water bottle to school every day.
- **58%** buy bottled water from the canteen at least twice a week.
- **62%** admit they often forget to carry a reusable bottle.
- **54%** believe buying bottled water is more convenient.
- **38%** think that their individual actions have little impact on the environment.
- **79%** said they would use reusable bottles more often if refill stations were easier to access.

Interview Responses

Student A

"I know plastic bottles are bad for the environment, but I usually forget my reusable bottle."

Student B

"Buying a bottle is faster and more convenient than carrying one."

Student C

"If there were more refill stations around the school, I would definitely use my reusable bottle."

Secondary Research

Research from environmental organisations reports that:

- Approximately **1 million plastic bottles are purchased every minute worldwide.**
- Many plastic bottles are **not recycled** and can remain in the environment for **hundreds of years.**
- Plastic pollution harms wildlife, especially marine animals, through ingestion and entanglement.
- Manufacturing plastic bottles requires large amounts of petroleum, water, and energy, increasing greenhouse gas emissions.
- Studies show that **convenience and habit** are among the biggest reasons why consumers continue to choose single-use plastics.

Your Task

Using the information provided, write **one PEEL paragraph (180–250 words)** answering the following question.

How does irresponsible consumer behaviour contribute to the excessive use of single-use plastic bottles, and why is this an important design problem?

Writing Guide

Point

State the design problem in one clear sentence.

Sentence starter:

One major design problem is that...

Evidence

Include:

- At least **two pieces of evidence** from the primary research.
- At least **one piece of evidence** from the secondary research.

Sentence starters:

- **According to the survey...**
- **Interview responses showed...**
- **Secondary research indicates...**

Explanation

Explain:

- Why this behaviour occurs.
- The causes of the problem.
- The effects on people and the environment.
- How the evidence supports your point.

Sentence starters:

- **This suggests that...**
- **One reason for this behaviour is...**
- **As a result...**
- **This demonstrates that...**

Link

Conclude by explaining why this problem matters.

Include:

- **Local impact** (school or community)
- **Global impact** (environment, climate, wildlife, waste)

Sentence starters:

- **Therefore...**
- **This problem affects our school because...**
- **On a global scale...**

Success Checklist

☐ I identified a clear design problem.

☐ I used evidence from the primary research.

☐ I used evidence from the secondary research.

☐ I explained the causes of the problem.

☐ I explained the effects of the problem.

☐ I linked the problem to both local and global impacts.

☐ I followed the PEEL structure throughout my paragraph.

Scenario

Greenfield Middle School has noticed that many students do not drink enough water during the school day. Although the school has installed water dispensers in different buildings, teachers have observed that students often wait until lunch or until they return home to drink water. Some students also forget to carry their water bottles or find them inconvenient to use during lessons.

The school wants to understand whether this is a genuine problem before deciding if a new product or solution should be designed.

Your task is to investigate the problem and justify whether there is a need for a solution.

Primary Research

Student Survey (30 students)

Question 1

How often do you refill your water bottle during school?

Response	Students
Never	8
Once	15
Twice or more	7

Question 2

Why do you not drink enough water at school? *(Students could choose more than one answer.)*

Response	Students
I forget to bring my bottle	12
My bottle becomes too heavy	10
Water dispensers are far away	18
I forget to drink	20
I don't feel thirsty	9

Question 3

When asked what would encourage them to drink more water, students said:

- "A reminder during the day."
- "A lighter bottle."
- "More water stations."
- "A bottle that tells me when to drink."
- "Something easy to carry."

Secondary Research

Source 1

Health experts recommend that children aged 11–14 drink approximately **1.5–2 litres of water each day** to stay hydrated.

Source 2

Even mild dehydration can reduce concentration, memory and learning performance during school activities.

Source 3

Research shows that students who drink enough water are generally more alert and able to participate actively in class.

Source 4

A study found that many middle school students forget to drink water because they become busy with classes, sports and social activities.

Source 5

Schools that increased access to drinking water reported that students refilled their bottles more often throughout the day.

Source 6

Designers often solve everyday problems by studying user habits before creating products that improve convenience and encourage positive behaviour.

Your Task

Using the **PEEL framework**, write **two well-developed paragraphs**.

Paragraph 1

Explain the problem using evidence from the primary and secondary research.

Paragraph 2

Justify why this problem needs a solution. Use evidence to explain the impact of the problem on students and link your ideas back to the need for a solution.

Success Criteria

Your response should:

- Clearly identify the problem.
- Use evidence from both the primary and secondary research.
- Explain how the evidence supports your point.
- Justify why the problem requires a solution.
- Organise ideas using the PEEL framework.
- Write entirely in your own words without copying the sources.

SA 1 Coursework :

1. A1. Aug 13 th - 9: 25 - Complete
2. A2. Aug 18 th 10:45 am - complete
3. A3 Aug 27 th - 10:05 - complete
4. A4 - Sept 3 - 10:55 - complete
5. B1 - Sept 8 - 11:35
6. B2 - Sept 15 - 11:35
7. B3 - Sept 22 - 10:55
8. B4.- Sept 29 - 12: 50